# Abhay Singh

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## EDUCATION

**Maharaja Agrasen Institute Of Technology** 2023 – 2027
*Bachelor of Technology in Computer Science Engineering (Data Science)* CGPA : 8.51/10
– **Coursework:** DBMS, Data Structures & Algorithms, Operating Systems, Machine Learning, Computer Networks, OOPS

## EXPERIENCE

**Web Development Intern** Jun 2025 – Jul 2025
*New Digital Bharat* Remote
– Developed and deployed a **responsive** web application using React, HTML, CSS, and JavaScript, improving page load performance by **30%** through optimized component rendering and asset compression.
– Built reusable UI components and implemented structured project architecture, reducing code redundancy by **25%** and improving **maintainability**.

## PROJECTS

**Hybrid AI Agent |** | *Python, LangGraph, LangChain, FastAPI, Streamlit, Pinecone*
– Developed a **hybrid RAG-based AI agent** using LangGraph and Groq, dynamically routing queries between internal knowledge retrieval and real-time web search via Tavily API.
– Built a **RAG pipeline** using Pinecone and Hugging Face sentence-transformer embeddings to retrieve relevant context and generate context-aware responses.
– Implemented **LLM-based RAG sufficiency evaluation** to assess retrieved context and trigger web search when internal knowledge was insufficient.
– Integrated **PDF ingestion and indexing** using PyPDFLoader, enabling document uploads, text chunking, embedding generation, and dynamic knowledge-base updates.
– Developed REST APIs with **FastAPI** and an interactive Streamlit interface featuring persistent conversation memory, user-controlled web search, and transparent agent execution traces.

**Penny Path | Expense Tracker** | *React, Node.js, Express, MongoDB, JWT, Bcrypt*
– Designed and deployed a **full-stack MERN** application for real-time expense tracking, supporting **CRUD operations** and handling multi-user transactions per session.
– Developed interactive financial dashboards using React and dynamic **data visualization** (Pie, Bar Charts), improving user insights and reducing manual tracking effort by **40%**.
– Implemented secure **JWT-based Authentication** with **bcrypt** password hashing and protected API routes to ensure secure session management.
– Built RESTful APIs and optimized MongoDB queries with indexing; enabled **Excel (.xlsx) export** functionality for offline data analysis.

## SKILLS

**Languages:** C,C++, Python, R, JavaScript
**Frameworks/Libraries:** React, Node.js, Scikit-Learn, FastAPI
**Generative AI & RAG:** Retrieval-Augmented Generation (RAG), LangChain, LangGraph, LLM Integration, Vector Search
**Cloud/Devops:** Docker, AWS **Databases:** MongoDB, MySQL, Pinecone
**Tools:** Git/GitHub, VS Code, Jupyter Notebooks, REST APIs
**Key Courses:** Operating Systems(OS), Database Management System(DBMS), Object Oriented Programming(OOPS)

## CERTIFICATES

**Google Prompt Design Course(Foundational):** Completed foundational training in crafting effective AI prompts to optimize large language model (LLM) responses.

**IBM Professional Certifications:** Completed Python for Data Science and Databases and SQL (51+ hours), covering advanced data analysis and database fundamentals.